KAMAL GUPTA

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Research Statement: Extending Large Multimodal Models into the physical dimension. My goal is to treat physical embodiment as a first-class modality – enabling AI systems to reason, plan, and act in the real world with the same fluidity as digital agents through long-range video understanding and robust post-training alignment

WORK EXPERIENCE

Tesla AI (Optimus), Staff Research Scientist — Palo Alto, May 2023 –

- **Embodied Alignment**: Training large-scale multimodal models with human-centric physical tasks, with Imitation Learning and RL from human demonstrations to achieve zero-shot transfer to the Optimus
- **Long-Range Video Understanding**: Developing architectures and training recipes to improve the temporal context window of vision-language models, enabling robots to reason over long-horizon tasks (minutes vs. seconds)
- **Physical Evals**: Developed rigorous "Physical Turing Test" evaluation suites to quantify model reasoning, causal understanding, and safety in unstructured human environments

Google Research, Research Intern — Mountain View, May 2022 – Nov. 2022

- **Spatial Alignment for Foundation Models**: Developed ASIC (Aligning Sparse in-the-wild Image Collections), a framework to bring order to unstructured visual data by establishing spatial and temporal consistency across disparate views (accepted to ICCV'23)

NVIDIA AI, Research Intern — Santa Clara, May 2021 – Nov. 2021

- **Discrete Latent Representations**: Engineered two-stage generative models using VQ-VAE style architecture to synthesize complex textured meshes, focusing on preserving physical and geometric constraints.

Amazon AWS, Research Intern — Pasadena, May 2019 – Aug. 2019

- **Foundational Generative Modeling**: Early research in adapting decoder-only autoregressive Transformer architectures (GPT-style) for structured visual domains (3D, documents, wireframes), demonstrating that "digital-first" architectures are highly effective for spatial reasoning (paper accepted in ICCV'21)

NetraDyne, Staff Research Engineer — Bengaluru, Mar. 2017 – Aug. 2018

Netradyne provides ADAS devices for commercial vehicles. I led a team of 4 to ship

- **Temporal Video Reasoning**: on-device models to interpret long-range temporal cues (distraction, drowsiness, and complex driving maneuvers) in varied environments, managing the "long-tail" of real-world physical data.
- **Large-Scale Behavioral Alignment**: a real-time safety pipeline to align driver behavior with human safety standards, processing millions of driving miles to refine edge-case detection.

Poolka AI, Cofounder, CTO — Bengaluru, Apr. 2016 – Feb. 2017

- **Full-stack Multimodal Model Deployment**: Designed and shipped Fairi, an early chatbot agent that integrated visual understanding (pose, segmentation) with language models to provide fashion recommendations.

American Express AI Labs, Research Engineer — Bengaluru, July 2013 – Mar. 2016

- Large scale recommendation systems, Geometric deep learning on financial data.
- **Platinum Genius Medal** - Systems and methods for customized real time data delivery — Dec. 2015
- **Trainer of the Quarter** - Hadoop course for >200 AmEx employees in NYC, Gurgaon, Bangalore — Dec. 2014

Robotics Institute, **Carnegie Mellon University**, June 2011 – June 2012 Research Intern, Pittsburgh, PA

- Developed an approach to predict vineyard yields automatically and non-destructively with cameras (web)
- Estimated global pose of MAV using stereo visual odometry fused with infrequent GPS measurements (web)

EDUCATION

University of Maryland, College Park — Aug. 2018 - May 2023

Ph.D. in Computer Science (4.0/4.0), Kulkarni fellow, Dean's fellow, Outstanding GRA Award

Thesis: Learning and Composing Primitives for the Visual World

Indian Institute of Technology Delhi — Aug. 2007 - Dec. 2012

Bachelors + Masters in Electrical Engineering (8.6/10.0)

Thesis: Pose Estimation for a Micro-Aerial Vehicle in GPS-denied environments

SELECTED PUBLICATIONS AND PATENTS

Visit Google Scholar tC3td8cAAAAJ for the complete list (⋆ denotes equal contribution)

- Measuring Style Similarity in Diffusion Models — *ECCV* 2024 [web, code]
  G. Somepalli, A. Gupta, **K. Gupta**, S. Palta, M. Goldblum, J. Geiping, A. Shrivastava, T. Goldstein
- LiFT: A Surprisingly Simple Lightweight Feature Transform for Dense ViT Descriptors — *ECCV* 2024 [web, code]
  S. Suri⋆, M. Walmer⋆, **K. Gupta**, A. Shrivastava
- SHACIRA: Scalable HAsh-grid Compression for Implicit Neural Representations — *ICCV* 2023 [web, code]
  S. Girish, A. Shrivastava, **K. Gupta**
- Chop & Learn: Recognizing and Generating Object-State Compositions — *ICCV* 2023 [web, code]
  N. Saini, H. Wang, A. Swaminathan, V. Jayasundara, B. He, **K. Gupta**, A. Shrivastava
- ASIC: Aligning Sparse in-the-wild Image Collections — *ICCV* 2023 [web, code]
  K. Gupta, V. Jampani, C. Esteves, A. Shrivastava, A. Makadia, N. Snavely, A. Kar
- Teaching Matters: Investigating the Role of Supervision in Vision Transformers — *CVPR* 2023 [web, code]
  M. Walmer⋆, S. Suri⋆, **K. Gupta**, A. Shrivastava
- LilNetX: Lightweight Networks with EXtreme Compression & Structured Sparsification — *ICLR* 2023 [web, code]
  S. Girish, **K. Gupta**, S. Singh, A. Shrivastava
- Neural Space-Filling Curves — *ECCV* 2022 [web, code]
  H. Wang, **K. Gupta**, L. Davis, A. Shrivastava
- PatchGame: Learning to Signal Mid-level Patches in Referential Games — *NeurIPS* 2021 [web, code]
  K. Gupta, G. Somepalli, Anubhav, V. Jayasundara, M. Zwicker, A. Shrivastava
- LayoutTransformer: Layout Generation and Completion with Self-attention — *ICCV* 2021 [web, code]
  K. Gupta, J. Lazarow, A. Achille, L. Davis, V. Mahadevan, A. Shrivastava
- Improved Modeling of 3D Shapes with Multi-view Depth Maps — *3DV* 2020 [web, code]
  K. Gupta⋆, S. Reddy⋆, K. Shah⋆, A. Shrivastava, M. Zwicker
- PatchVAE: Learning Local Latent Codes for Recognition — *CVPR* 2020 [web, code]
  K. Gupta, S. Singh, A. Shrivastava
- Applying Multi-Dimensional Variables to Determine Fraud — USPTO 16/426826, 2019
  K. Gupta, V. Jain
- Systems and methods for Updating Fraud Detection Variable — USPTO 15/258880, 2018
  K. Gupta, V. Jain
- Systems and methods for customized real time data delivery — USPTO 14/961614, 2015
  S. Sanyal, S. Purkayastha, T. Choudhuri, A. Choudhary, V. Grover, M. Naeem, P. Mehta, **K.Gupta**, A. Agarwal

VOLUNTEER INITIATIVES

- **Graduate Student Mentor**: CVPR Academy for first-time CVPR attendees — 2022
- **Co-organizer:** SIGGRAPH RCDC mentorship program for Graphics graduate school applicants — 2021
- **Instructor**: AI4ALL summer program for high school students — 2020
- **Reviewer**: Computer Graphics - SIGGRAPH 2022, SIGGRAPH Asia 2023. Computer Vision - ECCV 2022. CVPR 2023, 2022⋆, 2021. ICCV 2021. Machine Learning - NeurIPS 2022, 2023. ICLR 2022. (⋆ denotes Outstanding Reviewer Award)

MISCELLANEOUS

- Kulkarni Research Fellowship (web) — 2020
- Dean's fellowship for outstanding academic achievement — 2018-20
- **Coordinator**: Robotics Club IIT Delhi — 2008-10
- Master's Research Scholarship by Ministry of Human Resources and Development, Govt. of India — May 2011
- CBSE Merit Scholarship for All India Rank 38 (600,000 appeared) in AIEEE, now called JEE-Main — 2007
- National top 1% (40,000 appeared) in Physics Olympiad and Chemistry Olympiad — 2006
- National Talent Search Exam Scholarship by Govt. of India — 2005